ECE 329 **Homework 1** **Due: Mon, Sep 22, 2025, 11:59 PM**

Totaling 50pts with 5pts extra Bonus Points.

1. (10pts) **Review of vectors:** Consider the 3D vectors

$$\begin{aligned} \mathbf{A} &= \hat{x} + \hat{y} - \hat{z}, \\ \mathbf{B} &= (1, -2, -3), \\ \mathbf{C} &= 2\hat{x} - \hat{y} + 3\hat{z}, \end{aligned}$$

where $\hat{x} \equiv (1, 0, 0)$, $\hat{y} \equiv (0, 1, 0)$, and $\hat{z} \equiv (0, 0, 1)$ constitute an orthogonal set of unit vectors directed along the principal axes of a *right-handed* Cartesian coordinate system. Vectors can also be represented in component form — e.g., $\mathbf{C} = (2, -1, 3)$, which is equivalent to $2\hat{x} - \hat{y} + 3\hat{z}$.

Determine:

a) The vector $\mathbf{D} = \mathbf{A} + \mathbf{B}$,

b) The vector $\mathbf{A} + \mathbf{B} - 4\mathbf{C}$,

c) The vector *magnitude* $|\mathbf{A} + \mathbf{B} - 4\mathbf{C}|$.

d) The unit vector $\hat{u}$ along vector $\mathbf{A} + 2\mathbf{B} - \mathbf{C}$.

e) The *dot product* $\mathbf{A} \cdot \mathbf{B}$.

f) The *cross product* $\mathbf{B} \times \mathbf{C}$.

2. **Electrostatic charges and fields:** Charges $Q_1 = 10^{-9}$ C and $Q_2 = -Q_1$ are located at points P_1 and P_2 having the position vectors $\mathbf{r}_1 = -\hat{z} = (0, 0, -1)$ m and $\mathbf{r}_2 = \hat{z} = (0, 0, 1)$ m, respectively.

   a) (10pts) Determine the static electric field vector $\mathbf{E}$ at points P_3 and P_4 having the position vectors $\mathbf{r}_3 = (0, 0, 0)$ m and $\mathbf{r}_4 = \hat{x} = (1, 0, 0)$ m, respectively, using the Coulomb's electric field formula and the approximation that $\epsilon_o \approx \frac{1}{36\pi \times 10^9}$ F/m. .

   b) (5pts) **Bonus problem:** Make a Mathematica (or other programing language) based 3D vector plot representing $\mathbf{E}$ described above in the xOz plane. (Hint see MP1 from Class Calendar ...) and copy/paste the code and the plot here.

3. (10pts) **Fields from forces - Lorentz force:** A particle with charge $q = 4$ C located at the origin $\mathbf{r} = (x, y, z) = 0$ of the lab frame at time $t = 0$ is observed to accelerate with forces

$$\mathbf{F}_1 = 2\hat{z}, \quad \mathbf{F}_2 = 0, \quad \mathbf{F}_3 = \hat{z}\,\mathrm{N}$$

if the velocity of the particle is

$$\mathbf{v}_1 = 0, \quad \mathbf{v}_2 = 2\hat{y}, \quad \mathbf{v}_3 = -\hat{x}\,\frac{\mathrm{m}}{\mathrm{s}},$$

respectively. Use the Lorentz force equation $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ to determine the fields $\mathbf{E}$ and $\mathbf{B}$ at the origin. **Hint:** assume $\mathbf{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$ and then solve the three vector equations obtained from $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ for the unknowns B_x, B_y, B_z, and $\mathbf{E} = E_x\hat{x} + E_y\hat{y} + E_z\hat{z}$.

4. (10pts) **Field transformations:** In the lab frame $\mathbf{E} = 4\hat{z}$ V/m, $\mathbf{B} = -2\hat{y}$ T, and a point charge $q = 1$ C is observed to be moving with velocity $\mathbf{v} = 2\hat{x}$ m/s at the instant $t = 0$.

   a) What is the electric field $\mathbf{E}'$ measured in the frame of reference of q? Determine $\mathbf{E}'$ by ensuring that the Lorentz force applied on charge q is identical in both reference frames.

   b) Is this charge being accelerated or not under the influence of the fields $\mathbf{E}$ and $\mathbf{B}$? Discuss.

5. (10pts) **Current density and surface integrals:** Let $\mathbf{J} = -z^2(\hat{x}+\hat{y}+\hat{z})$ A/m^2 denote the electrical *current density* field — i.e., current flux per unit area — in a region of space represented in Cartesian coordinates. A current density of $\mathbf{J} = -z^2(\hat{x}+\hat{y}+\hat{z})$ A/m^2 implies the flow of electrical current in the direction $\frac{\mathbf{J}}{|\mathbf{J}|} = -\frac{\hat{x}+\hat{y}+\hat{z}}{\sqrt{3}}$ with a magnitude of $|\mathbf{J}| = z^2\sqrt{3}$ amperes (A) per unit area.

   a) Calculate the total **current flux** $\oint_S \mathbf{J}\cdot d\mathbf{S}$ "out" of a closed surface S enclosing a cubic volume $V = 1\,\mathrm{m}^3$ with vertices at $(x,y,z) = (0,0,0)$ and $(1,1,1)$ m.

   Hint: Surface S of cube V consists of six surfaces of square shapes having equal areas $S_i = 1\,\mathrm{m}^2$, $i = 1,2,\cdots,6$. The flux $\oint_S \mathbf{J}\cdot d\mathbf{S}$ is therefore the sum of six *surface integrals* $\int_{S_i} \mathbf{J}\cdot d\mathbf{S}$ taken over surfaces S_i, where the infinitesimal area vectors $d\mathbf{S}$ are, in turn, $\pm\hat{z}dxdy$, $\pm\hat{x}dydz$, and $\pm\hat{y}dzdx$ — by convention $d\mathbf{S}_i$ are taken as vectors pointing away from volume V (at each subsurface S_i) in flux calculations.

   b) Given the result of part (a), do you expect the total amount of electrical charge Q_V contained in volume V to increase or decrease? In answering this question assume that electric charge is *conserved* (as in real life!). Remember that electrical current represents electrical charges in motion.

   c) Given the result of part (b), do you expect the electric field flux $\oint_S \mathbf{E}\cdot d\mathbf{S}$ out of the same closed surface S to increase or decrease as a function of time? Explain in terms of Gauss' law.

Bonus problems will be rewarded separately and will be used at the end of the semester to "fill in" the "missing points" in the total HW score. Bonus problems are not required!